

past paper 2017

(Short Questions)

Q2(ii)

Find the solution of the following differential equations.

$$x \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 0$$

soln: Multiply 'x' on both sides

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$$

$$(x^2 D^2 - xD)y = 0$$

put

$$xD = \Delta$$

$$x^2 D^2 = \Delta(\Delta - 1)$$

$$\text{so, } (\Delta^2 - \Delta - \Delta)y = 0$$

$$(\Delta^2 - 2\Delta)y = 0$$

Auxiliary eq is

$$m^2 - 2m = 0$$

$$m(m - 2) = 0$$

$$m = 0, m = 2$$

$$y_c = C_1 e^{0x} + C_2 e^{2x}$$

$$y_c = C_1 + C_2 e^{2x}$$

Q2 (iii)

Determine the appropriate form for a particular solution of the following fifth order differential equation.

$$(D-2)^3(D^2+9)y = x^2 e^{2x} + x \sin 3x$$

Soln:

$$(D-2)^3(D^2+9)y = x^2 e^{2x} + x \sin 3x \quad \text{--- (i)}$$

For y_c

$$(D-2)^3(D^2+9)y = 0$$

Auxiliary eq is

$$(m-2)^3(m^2+9) = 0$$

$$m = 2, 2, 2, \quad m = 3i, -3i$$

so,

$$y_c = (C_1 + C_2 x + C_3 x^2) e^{2x} + e^{3ix} (C_4 \cos 3x + C_5 \sin 3x)$$

$$y_c = (C_1 + C_2 x + C_3 x^2) e^{2x} + C_4 \cos 3x + C_5 \sin 3x$$

For y_p from eq (i)

$$y_{p1} = x^k (A + Bx + Cx^2) e^{2x}$$

$$= x^3 (A + Bx + Cx^2) e^{2x}$$

$$y_{p1} = (Ax^3 + Bx^4 + Cx^5) e^{2x}$$

$$y_{p2} = x^k (E \cos 3x + F \sin 3x)$$

$$y_{p2} = x^2 [E \cos 3x + x^2 F \sin 3x]$$

$$y_p = (Ax^3 + Bx^4 + Cx^5) e^{2x} + x^2 [E \cos 3x + x^2 F \sin 3x]$$

Solve $Q2(iv) \quad \frac{y}{x^2} \frac{dy}{dx} + e^{2x^3+y^2} = 0$

soln: $\frac{y}{x^2} \frac{dy}{dx} + e^{2x^3} \cdot e^{y^2} = 0$

$$\frac{y}{x^2} \frac{dy}{dx} = -e^{2x^3} \cdot e^{y^2}$$

$$\frac{y}{e^{y^2}} dy = -x^2 e^{2x^3} dx$$

$$y e^{-y^2} dy + x^2 e^{2x^3} dx = 0$$

Integrate both sides

$$-\frac{1}{2} \int -2y e^{-y^2} dy + \frac{1}{6} \int 6x^2 e^{2x^3} dx = 0$$

$$-\frac{1}{2} [e^{-y^2}] + \frac{1}{6} [e^{2x^3}] = 0$$

$$-\frac{e^{-y^2}}{2} + \frac{e^{2x^3}}{6} = 0$$

$$\frac{e^{2x^3}}{6} - \frac{e^{-y^2}}{2} = 0 \Rightarrow \frac{e^{2x^3}}{6} = \frac{e^{-y^2}}{2}$$

Solve the following differential equation. $Q2(v)$

$$(D^2 + 6D + 13)^2 y = 0$$

soln:

$$(D^2 + 6D + 13)y = 0$$

Auxiliary eq. is

$$m^2 + 6m + 13 = 0, \quad m^2 + 6m + 13 = 0$$

solve by using quadratic formula

$$m = \frac{-6 \pm \sqrt{36 - 52}}{2}, \quad m = \frac{-6 \pm \sqrt{36 - 52}}{2}$$

$$m = \frac{-6 \pm 4i}{2}, \quad m = \frac{-6 \pm 4i}{2}$$

$$m = -3 \pm 2i, \quad m = -3 \pm 2i$$

$$y_c = e^{-2x} [(A+Bx)\cos 3x + (C+Ex)\sin 3x]$$

Long Questions

Q3

Solve the following differential equation.

$$xy'' - y' + \frac{y}{x} = x^3$$

soln:

Multiply by 'x' on both sides

$$x^2 y'' - xy' + y = x^3$$

$$(x^2 D^2 - xD + 1)y = x^3$$

For y_c

$$(x^2 D^2 - xD + 1)y = 0$$

put $x = e^t \Rightarrow \ln x = t$

put $x D = \Delta$

$x^2 D^2 = \Delta(\Delta-1)$

so, $(\Delta^2 - \Delta - \Delta + 1) = 0$

$(\Delta^2 - 2\Delta + 1) = 0$

Auxiliary eq is

$m^2 - 2m + 1 = 0$

$m^2 - m - m + 1 = 0$

$m(m-1) - 1(m-1) = 0$

$(m-1)(m-1) = 0$

$m = 1, 1$

$y_c = (C_1 + C_2 t) e^t$

For y_p

$y_p = \frac{1}{\Delta^2 - 2\Delta + 1} e^{3t}$

$y_p = \frac{1}{(\Delta-1)^2} e^{3t}$

$= \frac{1}{(3-1)^2} e^{3t}$

$y_p = \frac{1}{4} e^{3t}$

$y = (C_1 + C_2 t) e^t + (1/4) e^{3t}$

$y = (C_1 + C_2 \ln x) x + \frac{1}{4} x^3$

$y = x C_1 + x C_2 \ln x + \frac{x^3}{4}$

Q4

Use reduction of order to find a second solution of differential equation.

$$(3x+1)y'' - (9x+6)y' + 9y = 0, \quad (i) y_1 = e^{3x}$$

soln: Divide $(3x+1)$ on both sides

$$y'' - \frac{(9x+6)}{3x+1}y' + \frac{9}{3x+1}y = 0$$

let $y^* = Vy_1$

$$y = Ve^{3x}$$

$$\frac{dy}{dx} = V'e^{3x} + 3Ve^{3x}$$

$$\frac{d^2y}{dx^2} = V''e^{3x} + 3V'e^{3x} + 3V'e^{3x} + 9Ve^{3x}$$

$$\frac{d^2y}{dx^2} = (V'' + 6V' + 9V)e^{3x}$$

put in eq (i)

$$(3x+1)(V'' + 6V' + 9V)e^{3x} - (9x+6)(V' + 3V)e^{3x} + 9Ve^{3x} = 0$$

$$(3V''x + 18V'x + 27Vx + V'' + 6V' + 9V)e^{3x} + (-9V'x + 27Vx - 6V' + 18V)e^{3x} + 9Ve^{3x} = 0$$

$$(3x+1)V'' + 9V'x = 0$$

let $V'' = u$

$$V' = u'$$

$$(3x+1)u' + 9xu = 0$$

$$u' + \frac{9x}{3x+1} u = 0$$

$$u' + (3+9x)u = 0$$

$$\frac{du}{u} + (3+9x)dx = 0$$

Integrate both sides

$$\ln u + 3x + \frac{9x^2}{2} = C$$

$$\ln u = -3x - \frac{9x^2}{2} + C$$

$$u = e^{C-3x-\frac{9x^2}{2}}$$

$$u = e^C \cdot e^{-3x-\frac{9x^2}{2}}$$

$$v' = C_1 e^{-3x-\frac{9x^2}{2}}$$

$$\frac{dv}{dx} = C_1 e^{-3x-\frac{9x^2}{2}}$$

$$dv = C_1 e^{-3x-\frac{9x^2}{2}} dx$$

Integrate both sides

$$v = C_1 e^{-3x-\frac{9x^2}{2}} (-3-9x)$$

$$\frac{y}{e^{3x}} = -(9x+3) e^{-\frac{9x^2}{2}-3x}$$

$$y = -(9x+3) e^{-\frac{9x^2}{2}}$$

$$y = -\frac{(9x+3)}{e^{\frac{9x^2}{2}}} \text{ Ans.}$$

Q5.

Find the orthogonal trajectories of the given family of curves.

$$x^2 + 4y + 1 + Ce^{2y} = 0$$

soln: Diff w.r.t 'x'

$$2x + 4 \frac{dy}{dx} + 2Ce^{2y} \frac{dy}{dx} = 0$$

$$2x + 4 \frac{dy}{dx} + 2Ce^{2y} \frac{dy}{dx} = 0$$

$$x + 2 \frac{dy}{dx} + Ce^{2y} \frac{dy}{dx} = 0$$

$$x + 2 \frac{dy}{dx} + (-x^2 - 4y - 1) \frac{dy}{dx} = 0$$

$$x + (-x^2 - 4y - 1) \frac{dy}{dx} = 0$$

$$x - (x^2 + 4y + 1) \frac{dy}{dx} = 0$$

$$(x^2 + 4y + 1) \frac{dy}{dx} = -x$$

$$\frac{dy}{dx} = \frac{-x}{x^2 + 4y + 1}$$

Now orthogonal trajectories

$$\frac{dy}{dx} = \frac{1}{x/(x^2 + 4y + 1)}$$

$$\frac{dy}{dx} = \frac{(x^2 + 4y + 1)}{x}$$

$$\frac{dy}{dx} = -x + \frac{4y}{x} + \frac{1}{x}$$

$$\frac{dy}{dx} + \frac{4y}{x} = \frac{1}{x} - x \quad \dots (i)$$

$$I.F = e^{\int \frac{4}{x} dx} = e^{\ln x^4} = x^4$$

Multiply x^4 by eq (i)

$$x^4 \frac{dy}{dx} + \frac{4x^4 y}{x} = x^3 - x^5$$

$$\frac{d}{dx} (x^4 y) = x^3 - x^5$$

Integrate both sides

$$x^4 y = \frac{x^4}{4} - \frac{x^6}{6} + C$$

Q6

Find the general solution of the following differential equation

$$\frac{d^3 y}{dx^3} + \frac{dy}{dx} = \operatorname{cosec} x$$

soln: $(D^3 + D)y = \operatorname{cosec} x$

For yc

$$(D^3 + D)y = 0$$

Auxiliary eq is

$$m^3 + m = 0$$

$$m(m^2 + 1) = 0$$

$$m = 0, \quad m^2 = -1 \Rightarrow m = \pm i$$

Example: 21 Solve the differential equation.

$$\frac{d^3 y}{dx^3} + \frac{dy}{dx} = \cos ex$$

soln:

$$(D^3 + D)y = \cos ex$$

For y_c

$$(D^3 + D)y = 0$$

Auxiliary eq is

$$m^3 + m = 0$$

$$m(m^2 + 1) = 0$$

$$m = 0, \quad m^2 + 1 = 0 \Rightarrow m = \pm i$$

$$y_c = C_1 + C_2 \cos x + C_3 \sin x$$

$$y_p = u_1 + u_2 \cos x + u_3 \sin x$$

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

$$y_p' = u_1 y_1' + u_2 y_2' + u_3 y_3' + u_1' y_1 + u_2' y_2 + u_3' y_3$$

$$\Rightarrow u_1' y_1 + u_2' y_2 + u_3' y_3 = 0 \quad \text{--- (i)}$$

$$y_p' = u_1 y_1' + u_2 y_2' + u_3 y_3'$$

$$y_p'' = u_1 y_1'' + u_2 y_2'' + u_3 y_3'' + u_1' y_1' + u_2' y_2' + u_3' y_3'$$

$$\Rightarrow u_1' y_1' + u_2' y_2' + u_3' y_3' = 0 \quad \text{--- (ii)}$$

$$y'' = u_1 y_1'' + u_2 y_2'' + u_3 y_3''$$

$$y''' = u_1 y_1''' + u_2 y_2''' + u_3 y_3''' + u_1' y_1'' + u_2' y_2'' + u_3' y_3''$$

We know

$$\frac{d^3 y}{dx^3} + p \frac{d^2 y}{dx^2} + Q \frac{dy}{dx} + R y = F(x)$$

So,

$$(u_1 y_1''' + u_2 y_2''' + u_3 y_3''' + u_1' y_1'' + u_2' y_2'' + u_3' y_3'') + p(u_1 y_1'' + u_2 y_2'' + u_3 y_3'') + Q(u_1 y_1' + u_2 y_2' + u_3 y_3') + R(u_1 y_1 + u_2 y_2 + u_3 y_3) = F(x)$$

$$u_1 (y_1''' + p y_1'' + Q y_1' + R y_1) + u_2 (y_2''' + p y_2'' + Q y_2' + R y_2) + u_3 (y_3''' + p y_3'' + Q y_3' + R y_3) + u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = F(x)$$

$$u_1(0) + u_2(0) + u_3(0) + u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = F(x)$$

So,

$$u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = \cos x \quad \text{--- (iii)}$$

So from (i), (ii) and (iii)

$$u_1 y_1 + u_2 y_2 + u_3 y_3 = 0$$

$$u_1(0) + u_2(0) + u_3(0) = 0$$

$$u_1'(0) + u_2'(0) + u_3'(0) = \cos x$$

or

$$u_1' + u_2' \cos x + u_3' \sin x = 0$$

$$u_1(0) + u_2'(\sin x) + u_3' \cos x = 0$$

$$u_1(0) - u_2' \cos x - u_3' \sin x = \cos x$$

$$\begin{bmatrix} 1 & \cos x & \sin x & | & 0 \\ 0 & -\sin x & \cos x & | & 0 \\ 0 & -\cos x & -\sin x & | & \cos x \end{bmatrix}$$

$$\begin{bmatrix} 1 & \cos x & \sin x & | & 0 \\ 0 & 1 & -\cot x & | & 0 \\ 0 & -\cos x & -\sin x & | & \cos x \end{bmatrix} \xrightarrow{-\frac{1}{\sin x} R_2}$$

$$R_1 - \cos x R_2, \quad R_3 + \cos x R_2$$

$$= \begin{bmatrix} 1 & 0 & \sin x + \frac{\cos^2 x}{\sin x} & | & 0 \\ 0 & -1 & -\cot x & | & 0 \\ 0 & 0 & -\sin x - \cos^2 x & | & \cos x \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \cos x & | & 0 \\ 0 & 1 & -\cot x & | & 0 \\ 0 & 0 & -\cos x & | & \cos x \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \cos x & | & 0 \\ 0 & 1 & -\cot x & | & 0 \\ 0 & 0 & 1 & | & -1 \end{bmatrix} \xrightarrow{\frac{1}{\cos x} R_2}$$

So

$$u_1 + \cos x u_3 = 0 \quad \text{--- (iv)}$$

$$u_2 - \cot x u_3 = 0 \quad \text{--- (v)}$$

$$u_3 = -1 \quad \text{--- (vi)}$$

From (vi)

$$u_3 = -1$$

Integrate

$$u_3 = -x$$

So

$$u_1 + \cos x (-1) = 0$$

$$u_1 = \cos x$$

Integrate

$$u_1 = \ln |\cos x - \cot x|$$

$$u_2 - \cot x (-1) = 0$$

$$u_2 = -\cot x$$

Integrate both

$$u_2 = -\ln |\sin x|$$

there $u_1 = \ln |\cos x - \cot x|, u_2 = \ln \sin x$

$$u_3 = -x$$

so,

$$y_p = \ln|\csc x - \cot x| - (\cos x) \ln|\sin x - x \sin x$$

Q14. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = \frac{2e^x}{x^2} \quad \text{--- (i)}$

soln: $(D^3 - 3D^2 + 3D - 1)y = \frac{2e^x}{x^2} \quad \text{--- (ii)}$

For y_c
 $(D^3 - 3D^2 + 3D - 1)y = 0$

Auxiliary eq. is

$$m^3 - 3m^2 + 3m - 1 = 0$$

solve by using synthetic division

$$\begin{array}{r|rrrrr} 1 & 1 & -3 & 3 & -1 & \\ & & \downarrow & & & \\ & 1 & -2 & 1 & 0 & \end{array}$$

$m=1$ are one root

Depressed eq. is

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m = 1, 1$$

$$y_c = (C_1 + C_2x + C_3x^2)e^x$$

$$y_c = C_1e^x + C_2xe^x + C_3x^2e^x$$

$$y_p = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p' = u_1'y_1 + u_2y_2' + u_3y_3' + u_1y_1' + u_2'y_2 + u_3'y_3$$

so,

$$u_1'y_1 + u_2'y_2 + u_3'y_3 = 0 \quad \text{--- (ii)}$$

$$y_p' = u_1y_1' + u_2y_2' + u_3y_3'$$

$$y_p'' = u_1y_1'' + u_2y_2'' + u_3y_3'' + u_1'y_1' + u_2'y_2' + u_3'y_3'$$

so,

$$u_1 y_1' + u_2 y_2' + u_3 y_3' = 0 \quad \text{--- (iii)}$$

$$y_p'' = u_1 y_1'' + u_2 y_2'' + u_3 y_3''$$

$$y_p''' = u_1 y_1''' + u_2 y_2''' + u_3 y_3''' + u_1' y_1'' + u_2' y_2'' + u_3' y_3''$$

put in eq (i)

$$(u_1 y_1''' + u_2 y_2''' + u_3 y_3''' + u_1' y_1'' + u_2' y_2'' + u_3' y_3'') - 3(u_1 y_1'' + u_2 y_2'' + u_3 y_3'') + 3(u_1 y_1' + u_2 y_2' + u_3 y_3') = \frac{2e^x}{x^2}$$

$$u_1 (y_1''' - 3y_1'' + 3y_1' - y_1) + u_2 (y_2''' - 3y_2'' + 3y_2' - y_2) + u_3 (y_3''' - 3y_3'' + 3y_3' - y_3) = \frac{2e^x}{x^2}$$

$$u_1(0) + u_2(0) + u_3(0) + u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = \frac{2e^x}{x^2}$$

$$u_1 y_1'' + u_2 y_2'' + u_3 y_3'' = \frac{2e^x}{x^2} \quad \text{--- (iv)}$$

$$u_1 e^x + u_2 x e^x + u_3 x^2 e^x = 0$$

$$u_1 e^x + u_2 (1+x) e^x + u_3 (2x+x^2) e^x = 0$$

$$u_1 e^x + u_2 e^x (2+x) + u_3 (2x+x^2) e^x = \frac{2e^x}{x^2}$$

$$= \begin{bmatrix} 1 & x & x^2 & | & 0 \\ 1 & (1+x) & 2x+x^2 & | & 0 \\ 1 & 2+x & x^2+4x+2 & | & 2/x^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & x & x^2 & | & 0 \\ 1 & 1+x & 2x+x^2 & | & 0 \\ 1 & 2+x & x^2+4x+2 & | & 2/x^2 \end{bmatrix}$$

$$\xrightarrow{R_2-R_1, R_3-R_1} \begin{bmatrix} 1 & x & x^2 & | & 0 \\ 0 & 1 & 2x & | & 0 \\ 0 & 2 & 4x+2 & | & 2/x^2 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 1 & x & x^2 & 0 \\ 0 & 1 & 2x & 0 \\ 0 & 0 & 2 & 2/x^2 \end{array} \right] R_3 - 2R_2$$

$$= \left[\begin{array}{ccc|c} 1 & x & x^2 & 0 \\ 0 & 1 & 2x & 0 \\ 0 & 0 & 1 & 1/x^2 \end{array} \right] \frac{1}{2} R_3$$

$$= \left[\begin{array}{ccc|c} 1 & x & 0 & -1 \\ 0 & 1 & 0 & -2/x \\ 0 & 0 & 1 & 1/x^2 \end{array} \right] \begin{array}{l} R_1 - x^2 R_3 \\ R_2 - 2x R_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2/x \\ 0 & 0 & 1 & 1/x^2 \end{array} \right] R_1 - x R_2$$

$$u_1' = 1, u_2' = -2/x, u_3' = 1/x^2$$

Integrate

$$\text{so, } u_1 = x, u_2 = \ln x^{-2}, u_3 = -1/x$$

$$y_p = x e^x + \ln x^{-2} (x e^x) + -\frac{1}{x} (x^2 e^x)$$

$$y_p = x e^x + x e^x \ln x^{-2} - x e^x$$

$$y_p = x e^x \ln x^{-2} \text{ Ans.}$$